

GAINES COUNTY AP, TX, USA

WMO: 722107

Lat: 32.675N

Lon: 102.654W

Elev: 1010

StdP: 89.76

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03056

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.1	-16.6	1.0	2.2	-14.5	1.2	5.3	13.0	12.4	11.7	13.4	2.8	40	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	38.2	18.2	37.1	18.4	35.8	18.5	21.8	29.5	21.3	29.0	20.8	28.5	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	16.9	23.3	19.7	16.3	23.0	18.9	15.5	22.8	68.9	29.6	66.8	28.7	65.1	28.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.2	8.2	-10.8	41.0	2.5	1.5	-12.7	42.1	-14.1	43.0	-15.6	43.8	-17.4	44.9		
			-11.7	22.7	2.4	0.9	-13.4	23.4	-14.8	23.9	-16.2	24.4	-17.9	25.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.2	8.4	13.0	17.0	21.5	26.7	27.0	26.7	22.8	17.2	11.0	6.5		
	DBStd	8.67	4.81	5.40	4.91	4.36	4.39	2.70	2.29	2.47	3.13	4.38	4.69	4.86		
	HDD10.0	427	136	83	27	5	1	0	0	0	0	6	43	128		
	HDD18.3	1589	377	277	173	73	21	0	1	0	5	73	221	367		
	CDD10.0	2996	17	39	119	216	358	501	527	516	383	227	74	19		
	CDD18.3	1115	0	1	7	34	120	251	269	258	137	37	2	0		
	CDH23.3	14002	6	48	249	697	1648	3159	3222	2985	1397	501	82	8		
	CDH26.7	7277	0	7	63	266	846	1834	1802	1643	636	167	14	0		
(14) Wind	WSAvg	3.7	3.8	4.0	4.2	4.6	4.2	4.0	3.2	3.0	3.0	3.3	3.4	3.7		
(15) Precipitation	PrecAvg	451	15	19	22	23	53	59	62	55	64	36	23	16		
	PrecMax	843	59	62	114	92	192	258	248	190	234	141	158	92		
	PrecMin	199	0	0	0	0	0	1	1	1	0	0	0	0		
	PrecStd	140	15	17	25	27	45	47	56	47	60	41	30	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	27.8	30.8	33.6	38.0	40.8	38.9	39.0	36.2	32.5	28.6	24.8		
		MCWB	10.1	11.9	12.8	14.0	16.1	17.4	18.9	18.9	18.8	16.5	13.1	10.5		
	2%	DB	21.8	24.9	28.5	31.9	36.0	38.8	37.4	37.3	34.2	30.6	25.9	22.0		
		MCWB	9.3	10.7	12.5	13.4	15.6	17.6	19.1	19.4	18.2	16.4	12.5	10.0		
	5%	DB	18.7	22.3	26.3	30.0	34.0	37.2	36.1	35.9	32.6	28.6	23.1	18.8		
		MCWB	8.0	9.8	11.6	13.0	15.2	17.9	19.2	19.3	18.1	15.7	11.6	8.7		
	10%	DB	16.0	19.3	23.8	27.7	31.8	35.4	34.8	34.4	30.7	26.3	20.8	15.8		
		MCWB	7.0	8.5	10.9	12.9	15.2	18.2	19.3	19.5	18.1	15.3	10.8	7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	13.5	16.7	18.2	20.7	21.9	22.9	22.7	21.9	20.8	15.9	13.2		
		MCDB	17.0	21.9	22.0	24.4	27.3	30.5	28.8	29.4	27.5	26.8	20.2	17.1		
	2%	WB	10.6	11.9	15.3	16.9	19.7	21.2	21.8	21.8	21.1	19.7	14.5	10.9		
		MCDB	18.5	20.9	20.7	22.8	25.6	29.8	29.1	30.1	26.8	24.9	20.1	18.8		
	5%	WB	8.9	10.6	13.9	15.7	18.8	20.6	21.1	21.3	20.4	18.5	13.2	9.7		
		MCDB	17.9	20.4	20.7	22.3	25.2	29.3	29.0	29.3	26.3	23.9	20.1	17.8		
	10%	WB	7.4	9.2	12.4	14.6	17.9	20.2	20.6	20.7	19.9	17.3	12.0	8.0		
		MCDB	15.6	18.6	21.1	22.1	24.8	28.8	28.9	28.6	25.9	22.3	19.2	15.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	15.7	16.6	17.0	15.5	15.5	13.9	14.0	13.3	14.8	15.5	14.4		
		MCDBR	20.0	20.6	20.6	20.2	19.2	18.1	16.1	16.1	16.7	17.7	19.2	19.7		
	5% WB	MCWBR	10.2	9.8	8.6	8.0	6.6	4.8	4.1	4.0	4.8	6.5	8.6	10.0		
		MCWBR	17.8	18.9	17.3	16.2	14.8	15.2	13.6	13.6	12.4	13.5	16.0	17.4		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.272	0.292	0.323	0.337	0.374	0.391	0.370	0.349	0.315	0.284	0.271		
	taud		2.571	2.543	2.499	2.404	2.429	2.377	2.372	2.402	2.434	2.506	2.527	2.538		
	Ebn at Noon		978	995	991	965	947	907	890	907	916	930	938	944		
	Edh at Noon		79	90	102	118	116	122	122	117	108	93	82	78		
(44) All-Sky Solar Radiation	RadAvg		3.49	4.32	5.60	6.90	7.39	7.74	7.30	6.76	5.63	4.59	3.64	3.07		
	RadStd		0.33	0.40	0.41	0.42	0.50	0.36	0.38	0.28	0.40	0.47	0.37	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page